Mahmoud Ahmed Elsayed

Data Engineer

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SUMMARY

Data engineer with an end-to-end Azure lakehouse pipeline (Data Factory, Databricks, Delta Lake, Synapse, Power BI), warehouse projects on Snowflake and dbt, and military-service Python ETL that turned tens of thousands of Word documents into queryable datasets (still in use after handover).

SKILLS

- **SQL & Databases:** T-SQL, Spark SQL, PL/SQL; SQL Server, MySQL, PostgreSQL, SQLite; stored procedures, dynamic SQL, window functions, CTEs
- **Data Modeling & Warehousing:** dimensional modeling, star & snowflake schemas, SCD Types 0–3, ERD & normalization, ETL/ELT design, Medallion Architecture, data cleaning
- **Python:** pandas, NumPy, PySpark, Matplotlib, Seaborn, python-docx
- **Azure:** Data Factory, Databricks, Synapse Analytics, Data Lake Storage Gen2 (ADLS), Key Vault
- **Data Platforms:** Delta Lake, Snowflake, dbt, SSIS
- **BI & Reporting:** Power BI, Tableau, Microsoft Excel
- **Tools:** Docker, Git & GitHub, Linux, Bash, Jupyter

CERTIFICATIONS

- **AWS Certified Cloud Practitioner** (CLF-C02) — Amazon Web Services, 2026 `verify`

EXPERIENCE

Office Data & Inventory Assistant (Military Service) — Jun. 2025 – Jun. 2026

- *Border Guards Intelligence & Security Office* — *Sallum, Egypt*

- Built a Python ETL pipeline (python-docx, pandas) that migrated tens of thousands of templated Word inventory documents spanning 2018–2026 into structured monthly and yearly datasets.
- Cut aggregate inventory questions from up to a week of error-prone manual cross-checking to direct queries; built quarterly inventory-comparison reports in Power BI.
- Developed reusable extraction tools that pull inventory tables and reporting dates from document context, eliminating repetitive manual re-entry.
- Normalized inconsistent, manually entered Arabic text with pandas so records matched reliably across years and categories.
- Consolidated output into 10 standardized, category-specific tables that feed a UI for browsing records and generating documents.
- Tools remain in use by non-technical staff; the unit requested a dedicated IT soldier specifically to maintain the system after the posting ended.

Freelance IT Tutor — Jan. 2025 – May 2025

- *Upwork* — *Remote*

- Tutored a US IT undergraduate in SQL, Python, and databases, two sessions per week across a full semester; the student passed his courses.

PROJECTS

NYC Taxi Lakehouse — Azure ELT Pipeline (GitHub) **Oct. 2024 – Nov. 2024**

◇ *DEPI Graduation Project*

- Led a team of 4 through design, build, and delivery of the end-to-end pipeline.
- Designed and built the Docker-hosted SQL Server source (ER model, 3NF schema), seeded with NYC Taxi plus synthetic Faker data; teammates orchestrated runs via an Apache Airflow DAG.
- Ingested data into ADLS Gen2 with Azure Data Factory and implemented Bronze/Silver/Gold Medallion layers in Databricks with PySpark and Delta Lake, including data cleansing and SCD Type 1 handling.
- Modeled the Gold layer as a star schema supporting reporting and a fare-prediction ML model.
- Queried Gold via Synapse serverless SQL and built a Power BI dashboard of trip and fare insights.

End-to-End Azure Data Engineering Pipeline (GitHub) **Sep. 2024 – Nov. 2024**

◇ *Personal Project*

- Built an ADF-orchestrated pipeline over AdventureWorksLT with credentials managed in Azure Key Vault.
- Transformed data bronze-to-silver-to-gold in Databricks notebooks; served via Synapse views and Power BI.

Airbnb ELT Pipeline — Snowflake & dbt (GitHub) **Jul. 2026 – Present**

◇ *Personal Project*

- Built an end-to-end ELT pipeline on Snowflake with dbt Core: RAW to Bronze to Silver to Gold medallion layers, with incremental loading between them.
- Implemented SCD Type 2 history through dbt snapshots, selecting a snapshot strategy that worked around missing source timestamps.
- Configured Snowflake access with key-pair authentication, scoped roles, and warehouse cost controls; documented the tradeoff behind each decision in the repo.

Multi-Source Data Warehouse Integration (GitHub) **Jun. 2024 – Jul. 2024**

◇ *Personal Project*

- Migrated data from two SQL Server databases into a staging schema using Azure Data Factory.
- Wrote dynamic stored procedures (sp_executesql) that generate surrogate-key mapping tables, preventing key collisions when merging records from the two sources.
- Loaded the integrated data from transformation views into conformed base tables via stored procedures.

EDUCATION

Digital Egypt Pioneers Initiative (DEPI) **Apr. 2024 – Oct. 2024**
◇ *Diploma in Data Engineering* *Alexandria, Egypt*

Alexandria University **2018 – 2024**
◇ *Bachelor of Engineering* *Alexandria, Egypt*

Information Technology Institute (ITI) **Jan. 2023 – May 2023**
◇ *Diploma in Business Intelligence* *Alexandria, Egypt*

Udacity **Sep. 2022 – Dec. 2022**
◇ *Nanodegree in Data Analysis* *Online*

COURSES

- **Azure Data Fundamentals (DP-900) course** – KodeKloud (2024)
- **Advanced SQL for Data Engineering** – 365 Data Science (2024)
- **Taming Big Data with Apache Spark** – Udemy (2024)
- **CS50's Introduction to Databases with SQL** – HarvardX (2024)
- **Data Analyst Career Track** – 365 Data Science (2024)
- **Docker Level 1** – KodeKloud (2024)